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Database Management Systems (DBMS) · Unit 1 · Introduction & ER Model · 2025

Overview

These notes cover **Introduction & ER Model**. Design, query and manage relational databases — from ER modelling and SQL to normalization, transactions and recovery. Each section states the core idea, a worked intuition and the points examiners look for.

1. Database System Architecture

A DBMS separates the logical view of data from physical storage using a three-schema architecture: external, conceptual and internal levels.

Why it matters: Database System Architecture appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of database system architecture in one line.
- A small example showing database system architecture in practice.
- Common mistakes students make and how to avoid them.

2. Data Independence

Logical data independence lets the conceptual schema change without rewriting applications; physical data independence hides storage changes from the conceptual level.

Why it matters: Data Independence appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of data independence in one line.
- A small example showing data independence in practice.
- Common mistakes students make and how to avoid them.

3. Entity-Relationship Model

ER diagrams model the real world as entities, attributes and relationships, and are the standard first step of database design.

Why it matters: Entity-Relationship Model appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of entity-relationship model in one line.
- A small example showing entity-relationship model in practice.
- Common mistakes students make and how to avoid them.

4. Cardinality and Participation

Mapping cardinalities (1:1, 1:N, M:N) and total or partial participation constraints describe how entities relate.

Why it matters: Cardinality and Participation appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of cardinality and participation in one line.
- A small example showing cardinality and participation in practice.
- Common mistakes students make and how to avoid them.

5. Weak Entity Sets

A weak entity has no primary key of its own and is identified through an identifying relationship with its owner entity.

Why it matters: Weak Entity Sets appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of weak entity sets in one line.
- A small example showing weak entity sets in practice.
- Common mistakes students make and how to avoid them.

6. Keys

Super keys, candidate keys, primary keys and foreign keys uniquely identify tuples and enforce referential integrity.

Why it matters: Keys appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of keys in one line.
- A small example showing keys in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Database System Architecture · Data Independence · Entity-Relationship Model · Cardinality and Participation · Weak Entity Sets · Keys